TECHNICAL NOTE

Governance costing model: technical note (v018)

Emanuel Marques Queiroga · 9 November 2026 · AI in Education Procurement Tool · v018

Summary

The governance cost is estimated option by option, with explicit calculations and parameters that users can see and adjust in the tool. Each option selected in the plan has its own calculation, and the total cost is the sum of the options plus three additional costs calculated on top of them.

Most of the cost comes from people's work. For 25 options, the tool uses a reference team, with the roles needed, the number of people, each person's time share and the months of work. The monthly cost of each role starts from the country's average earnings for the corresponding occupational group (ILOSTAT, ILO), updated to 2026 with the country's inflation (IMF) and increased by the mandatory employer contribution (SSA and ISSA). Roles that require AI, data or digital security skills receive a 28% wage premium, measured across 1.3 billion job postings (Lightcast, 2025).

Travel and in-person activities (25%), overhead (15%) and an administrative fee (7%) are applied to the cost of the options, each with a public source and a question that indicates whether it applies to the plan. The tool shows the economic cost of the plan and, separately, the new outlay, which excludes the work of existing civil servants.

Scope and cost structure

Governance has 3 challenges, 9 categories and 32 options, and each option has a calculation defined by its cost type. Categories 2.1 and 3.2 have no options, so the category itself is the cost line.

Cost type	Where it applies	Calculation	Source of the unit value
Reference team	25 options in categories 1.1, 1.2, 1.3, 2.1, 2.3 and 3.3	Sum over roles: people × time share × months × monthly cost	Country earnings (ILOSTAT) × inflation (IMF) × contributions (SSA and ISSA) × specialist premium, where applicable
Full-time staff	4 options in category 3.1	People × monthly cost × plan duration	Country earnings, group 2, with the same treatment; premium in the AI, data and IT options
Training	Category 3.2	Participants × hours × cost per hour	Reference estimate of US$12 per participant-hour
Entered amount	4 options in category 2.2	Total amount of the quotation or contract	Entered by the user
Percentage	Travel, overhead and administrative fee	Percentage × base	Global Fund, 2 CFR 200.414, UNSDG policy

General equation and notation

The total cost is the direct cost plus the three additional costs. The direct cost includes only the options that already have an amount; the others are shown as "missing amount" and enter the total once they are priced.

$$C = D + V + O + T, \qquad D = \sum_{o} C_o$$

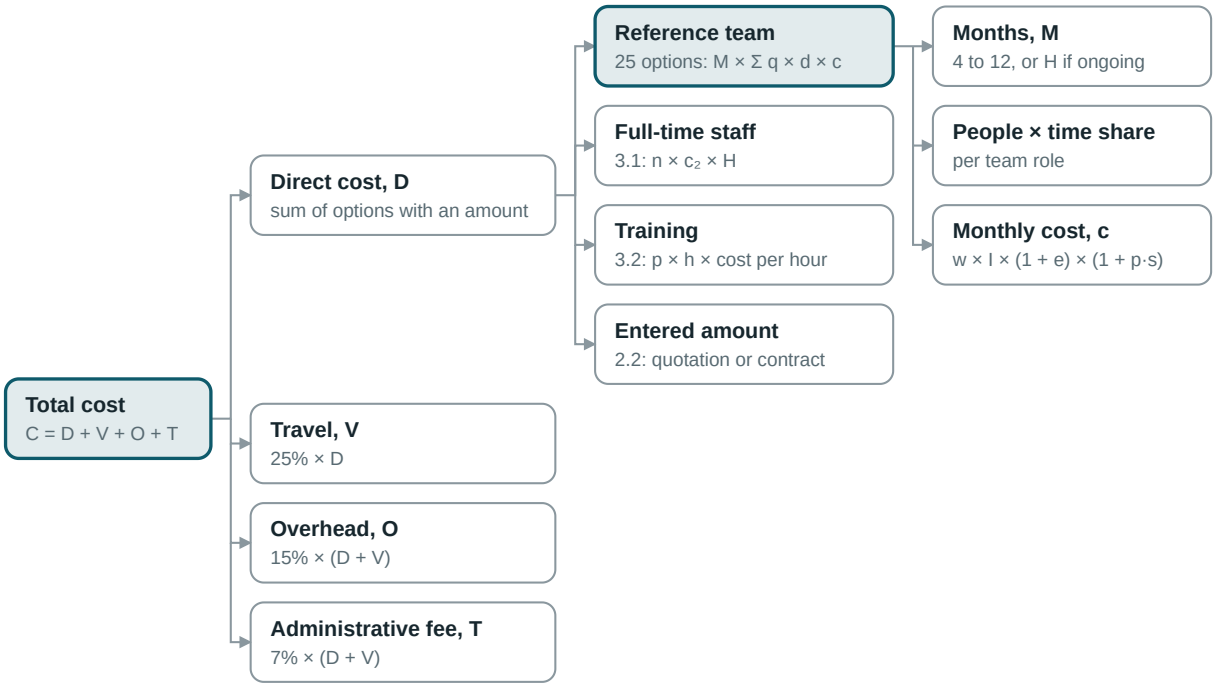


Figure 1. Structure of the governance cost, from the total to the parameters.

Symbol	Meaning	Unit
C	Total cost of the governance plan	plan currency
D	Direct cost: sum of the options that already have an amount	plan currency
C_o	Cost of option o	plan currency
V, O, T	Travel and in-person activities, overhead and administrative fee	plan currency
H	Plan duration (default 12)	months
M	Working months of the option (one-off) or H (ongoing)	months
q, d	People and time share of a team role	people; share of time
w	Average monthly earnings of the occupational group in the country, in the data year	plan currency per month
I	Inflation update factor, from the data year to 2026	index
e	Employer contribution on payroll	share
p, s	Specialist premium and specialist-role indicator (1 or 0)	share; 0 or 1
x	Exchange rate: plan currency per US$	currency per US$

Options with a reference team

The cost of an option with a reference team is the team's monthly cost multiplied by the months of work. The monthly cost adds up, role by role, the number of people, the time share and the monthly cost of one person in that role.

$$C_o = M \sum_r q_r \, d_r \, c_r, \qquad c_r = w_{g(r)} \cdot I \cdot (1+e) \cdot (1 + p\, s_r)$$

Months of work

One-off options, such as a regulation, a guideline or a sandbox, have their own reference months, from 4 to 12. Ongoing options, such as supervision, committees, interministerial coordination and data management, last the whole plan, so M = H.

Country earnings

Earnings w are the monthly average of employees in the role's ISCO-08 major group (1, managers; 2, professionals; 3, technicians and associate professionals; 4, clerical support workers), from the ILOSTAT indicator, for both sexes and the latest year available since 2015. When the plan uses the country's currency, the tool uses the local-currency value; otherwise, it converts the US$ value published by the ILO with the exchange rate x.

For countries without their own data, w is the US$ median of countries with data in the same World Bank region and income group, with at least three countries, or the median of the income group. In the absence of any estimate, the world median of countries with their own data is used. Values below US$30 a month are discarded as recording errors.

Inflation update

Because earnings data refer to different years, each value is updated from the data year to 2026 by the product of annual consumer price inflation in the following years, according to the IMF World Economic Outlook (with an estimate for 2025 and a projection for 2026). Earnings in local currency are updated with the country's inflation; earnings converted from US$, with United States inflation.

$$I = \prod_{t=a+1}^{2026} (1 + \pi_t)$$

Employer contributions

The factor 1 + e includes the mandatory employer social security contribution, as a percentage of payroll, from table 4 of Social Security Programs Throughout the World (SSA and ISSA; Africa and the Americas, 2019; Europe and Asia and the Pacific, 2018). Additional benefits set by contract or by specific legislation can be included by the user in the monthly cost of each role.

Specialist premium

Professionals with AI, data and digital security skills earn above the average of their occupational group. The model applies to these roles (s = 1, marked with an asterisk in the team table) a premium p of 28%, the salary difference between postings that do and do not require AI skills across 1.3 billion job ads (Lightcast, 2025). This is a conservative value: PwC (2025) estimated an average premium of 56% for jobs requiring AI skills, compared with 25% the year before. The percentage can be adjusted in the tool.

Editable values

In the tool, users can change the months of a one-off option, the number of people in each role, the monthly cost of each role and the specialist premium. An entered value replaces the estimate, and the line moves from "estimated" to "entered" once all its values have been entered.

Other options

Options without a reference team use simpler calculations, in which the quantity is always entered by the user. In the four full-time staff options (3.1), the cost is the number of people n times the monthly cost of group 2, calculated as above, times the plan duration; the AI, data and IT options receive the specialist premium.

$$C_o = n \cdot w_2 \cdot I \cdot (1 + e) \cdot (1 + p\, s_o) \cdot H$$

For staff training (3.2), the cost is the number of participants times the hours per participant times the cost per hour, whose reference estimate is US$12 per participant-hour, converted with the exchange rate. The value may vary with the delivery model and should be replaced by the contract value when available. For the digital infrastructure and hosting options (2.2), the cost is the total amount of the quotation or contract, because it depends on the provider, the data volume and the procurement model.

$$C_o = \text{participants} \cdot \text{hours} \cdot 12\, x$$

Additional costs

The three additional costs are percentages of the direct cost. Travel applies to D; overhead and the administrative fee apply to D + V, and neither enters the other's base.

$$V = p_V\, D, \qquad O = p_O\, (D + V), \qquad T = p_T\, (D + V)$$

Cost	Default	Base	Source of the default
Travel and in-person activities (fares, per diems, workshops, technical meetings)	25%	D	Global Fund: travel-related costs are 24.7% of programme budgets in cycle 7, excluding health products (27.6% in cycle 6; 28.2% in cycle 5)
Overhead (premises, IT, HR, accounting, legal)	15%	D + V	United States, 2 CFR 200.414(f): up to 15% of direct costs without a negotiated rate (2024 revision)
Administrative fee (charged by whoever executes the funds)	7%	D + V	UNSDG policy for UN pooled funds: 7% of programme costs

Each additional cost has a question that indicates whether it applies to the plan. Without an answer, the default applies; the answer that excludes the cost sets its percentage to zero.

Question	Answer that excludes the cost	Effect
Will the plan have technical meetings, workshops or travel?	No	p_V = 0; overhead and fee then apply to D only
Does the institution already cover indirect costs?	Yes, it already does	p_O = 0
Who will execute the funds?	The agency itself	p_T = 0; with a UN agency, 7%; with a support foundation, the contract fee (in Brazil, up to 15%, Decree 9,283/2018)

Plan cost and new outlay

The plan cost measures the value of all the work required, including that of people who are already civil servants of the agency. To support budget planning, users can mark, in each team, the roles held by existing civil servants, and the tool also shows the new outlay.

$$\text{New outlay} = C - \sum_{r \in \text{existing}} M\, q_r\, d_r\, c_r$$

Duration, exchange rate and totals

Governance has a single period of duration H (default 12 months, editable), and all costs are added up within it. A one-off option counts all its reference months, even if they exceed H, and the implementation roadmap flags these cases in its calendar.

The exchange rate x is the one entered in the case or, failing that, the ExchangeRate-API reference rate (USD base, 3 October 2026), rounded to four decimals so that the calculation can be checked by hand. Amounts in different currencies are never added together. Each line has one of three statuses, and the total shows the first two separately.

Status	When	In the total
Entered	All the line's values were entered by the user	Yes
Estimated	At least one value comes from an estimate (country earnings, reference team, default percentage)	Yes, identified as estimated
Missing amount	A quantity, price or months value is missing	No; shown as pending

Reference team parameters

The 25 teams have 2 to 8 people; one-off options last 4 to 12 months and ongoing ones the whole plan. The asterisk marks roles with the specialist premium, and the last column names the public case used to check the order of magnitude.

Cat.	Option	Months	People	Full-time equivalent	Team: role (profile) people × time share	Comparison
1.1	AI safety and risk classification	8	7	5.5	Coordination (G1) 1 × 0.5; Digital law and regulation specialist* (G2) 2 × 1; AI and risk assessment specialist* (G2) 1 × 1; Education policy specialist (G2) 1 × 0.5; Public consultation analyst (G3) 1 × 1; Administrative support (G4) 1 × 0.5	
1.1	Children's rights and protection	6	6	3.5	Coordination (G1) 1 × 0.5; Children's rights specialist (G2) 1 × 1; Child development and education specialist (G2) 1 × 0.5; AI specialist* (G2) 1 × 0.5; Public consultation analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.5	
1.1	Personal data protection and privacy	8	7	5.5	Coordination (G1) 1 × 0.5; Data protection law specialist* (G2) 2 × 1; Education data specialist* (G2) 1 × 1; Data protection officer* (G2) 1 × 0.5; Public consultation analyst (G3) 1 × 1; Administrative support (G4) 1 × 0.5	Kenya data protection office
1.1	Algorithmic fairness, non-discrimination, transparency, explainability and accountability	9	8	6.5	Coordination (G1) 1 × 0.5; AI ethics and fairness specialist* (G2) 1 × 1; Data scientist* (G2) 1 × 1; AI explainability specialist* (G2) 1 × 1; Fundamental rights specialist (G2) 1 × 1; Inclusive education specialist (G2) 1 × 0.5; Technical analyst (G3) 1 × 1; Administrative support (G4) 1 × 0.5	
1.1	Cybersecurity and incident reporting	6	6	4.5	Coordination (G1) 1 × 0.5; Information security specialist* (G2) 2 × 1; Digital law specialist* (G2) 1 × 0.5; Incident response analyst* (G3) 1 × 1; Administrative support (G4) 1 × 0.5	
1.1	Intellectual property and copyright	5	5	2.5	Coordination (G1) 1 × 0.25; Intellectual property specialist (G2) 1 × 1; Educational resources specialist (G2) 1 × 0.5; Generative AI specialist* (G2) 1 × 0.25; Administrative support (G4) 1 × 0.5	
1.2	Controlled testing environments	12	8	6.5	Coordination (G1) 1 × 1; AI specialist* (G2) 1 × 1; Regulation specialist (G2) 1 × 1; Educational assessment specialist (G2) 1 × 0.5; IT infrastructure and security specialist* (G2) 1 × 0.5; Pilot school liaison (G3) 1 × 1; Technical analyst (G3) 1 × 1; Administrative support (G4) 1 × 0.5	Norway sandbox
1.2	Regulatory supervision	ongoing	7	7	Coordination (G1) 1 × 1; AI supervision specialist* (G2) 3 × 1; Administrative law specialist (G2) 1 × 1; Technical analyst (G3) 1 × 1; Administrative support (G4) 1 × 1	AI Act, COM(2021) 206
1.2	Risk assessments	6	4	2.25	AI risk assessment specialist* (G2) 1 × 1; Data scientist* (G2) 1 × 0.5; Data protection specialist* (G2) 1 × 0.25; Technical analyst (G3) 1 × 0.5	Norway sandbox
1.3	Ethical AI guidelines for education	5	5	3	Coordination (G1) 1 × 0.25; AI ethics specialist* (G2) 1 × 1; Education specialist (G2) 1 × 1; Technical writer (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	AI procurement and vendor evaluation guidelines	5	6	3.5	Coordination (G1) 1 × 0.25; Public procurement specialist (G2) 1 × 1; AI specialist* (G2) 1 × 1; Legal specialist (G2) 1 × 0.5; Technical analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	Age-appropriate AI use guidelines	4	5	2.5	Coordination (G1) 1 × 0.25; Child development specialist (G2) 1 × 1; Education specialist (G2) 1 × 0.5; AI specialist* (G2) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	Teacher and student AI-use policies	4	6	2.75	Coordination (G1) 1 × 0.25; Education specialist (G2) 1 × 1; Assessment and academic integrity specialist (G2) 1 × 0.5; Legal specialist (G2) 1 × 0.25; Technical analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	AI risk management standards	5	5	3	Coordination (G1) 1 × 0.25; Risk management specialist (G2) 1 × 1; AI specialist* (G2) 1 × 1; Technical analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	AI transparency and documentation standards	4	5	2.5	Coordination (G1) 1 × 0.25; AI specialist* (G2) 1 × 1; Legal specialist (G2) 1 × 0.5; Technical analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	

Cat.	Option	Months	People	Full-time equivalent	Team: role (profile) people × time share	Comparison
1.3	Data governance and security standards	5	5	3	Coordination (G1) 1 × 0.25; Data governance specialist* (G2) 1 × 1; Information security specialist* (G2) 1 × 1; Technical analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	AI quality assurance and evaluation guidelines	5	5	3	Coordination (G1) 1 × 0.25; Educational assessment specialist (G2) 1 × 1; AI specialist* (G2) 1 × 1; Data analyst* (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
2.1	Data governance, protection and digital public infrastructure	ongoing	7	6.5	Coordination (G1) 1 × 1; Data governance specialist* (G2) 2 × 1; Data and interoperability architect* (G2) 1 × 1; Data analyst* (G3) 2 × 1; Administrative support (G4) 1 × 0.5	Kenya data protection office
2.3	Open-source adoption and maintenance	ongoing	4	3.5	Technical lead* (G1) 1 × 0.5; Developer* (G2) 2 × 1; Support technician (G3) 1 × 1	
2.3	Technology transfer	8	6	3.75	Coordination (G1) 1 × 0.5; IT specialist* (G2) 2 × 1; Contract manager (G2) 1 × 0.5; Technical trainer (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
2.3	Local customisation	8	6	5	Coordination (G1) 1 × 0.5; Developer* (G2) 2 × 1; Curriculum specialist (G2) 1 × 1; Language and translation specialist (G2) 1 × 0.5; Tester (G3) 1 × 1	
2.3	Vendor dependency reduction	4	4	2.25	Coordination (G1) 1 × 0.25; Public procurement specialist (G2) 1 × 1; IT architect* (G2) 1 × 0.5; Legal specialist (G2) 1 × 0.5	
3.3	Governance committees	ongoing	2	1	Executive secretary (G2) 1 × 0.5; Administrative support (G4) 1 × 0.5	
3.3	Interministerial coordination	ongoing	3	2.5	Coordination (G1) 1 × 1; Public policy analyst (G2) 1 × 1; Administrative support (G4) 1 × 0.5	
3.3	Institutional governance structures	ongoing	8	8	Coordination (G1) 1 × 1; AI specialist* (G2) 1 × 1; Data specialist* (G2) 1 × 1; Legal specialist (G2) 1 × 1; Education specialist (G2) 1 × 1; Technical analyst (G3) 2 × 1; Administrative support (G4) 1 × 1	Kenya data protection office

Each team was designed the way a coordinator would build it, with leadership, the specialists for each area and support staff. The supervision unit sits at the lower end of the range the European Commission estimated for the AI Act (1 to 25 full-time staff per country), and the sandbox and risk assessments follow the pace of Norway's sandbox (four projects a year, three to six months each).

Treatment of uncertainty

The model includes adjustments for the main sources of uncertainty in the estimate. The gap between the occupational group's average earnings and those of specialists is handled by the specialist premium; the age of the earnings data, by the inflation update; and the lack of national data, by ILOSTAT regional and world medians, always with the origin shown in each line's calculation.

Team composition, time shares and months were defined for each type of work and compared with public cases; these values may vary slightly with the context and can be adjusted role by role. When the same person coordinates more than one option, users only need to adjust the number of people or the coordinator's time share in each team. The sensitivity to the specialist premium in the Brazil test case is shown in the table below.

Specialist premium	Known cost of the test case (R$)	Change relative to the default
0%	5,182,500.93	−11.6%
25% (PwC, 2024 edition)	5,792,111.70	−1.2%
28% (default; Lightcast, 2025)	5,865,266.73	—
56% (PwC, 2025)	6,548,029.66	+11.6%

Confidence grades

Parameter	Grade	Source
Earnings by occupational group, country with data	A	ILOSTAT, by country and year
Consumer price inflation	A	IMF, World Economic Outlook
Employer contributions	A	SSA and ISSA, table 4
Exchange rate	A	ExchangeRate-API or entered in the case
Earnings for a country without data	B	ILOSTAT regional, income-group or world median
Specialist premium	B	Lightcast (2025); compared with PwC (2025)
Travel, overhead and fee percentages	B	Global Fund, 2 CFR 200.414, UNSDG
Team composition, time shares and months	C	Tool estimate, compared with public cases
Training cost per hour	C	Reference estimate, replaceable by the contract value

Worked example: Brazil test case

In the test case (Brazil, Brazilian reais, 12 months, 12 cost lines), the known cost is R$5,865,266.73: R$3,846,076.54 in options and R$2,019,190.19 in additional costs; two options are still without an amount. Brazil has its own ILOSTAT data (Continuous PNAD, 2025), updated to 2026 with the projected inflation of 4.4% (factor 1.044), and an employer contribution of 21%.

Group	Average monthly earnings, w (R$, 2025)	Monthly cost, c (R$)	Specialist monthly cost (R$)
G1 Managers	8,000.59	10,106.67	12,936.53
G2 Professionals	6,551.79	8,276.48	10,593.90
G3 Technicians and associate professionals	4,332.05	5,472.42	7,004.70
G4 Clerical support workers	2,762.83	3,490.12	4,467.35

The option "AI safety and risk classification" has 8 months and six roles, two of them with the specialist premium. The team's monthly cost is R$48,190.75, and the option costs R$385,526.04.

$$\text{monthly cost} = 0.5 \times 10{,}106.67 + 2 \times 10{,}593.90 + 10{,}593.90 + 0.5 \times 8{,}276.48 + 5{,}472.42 + 0.5 \times 3{,}490.12 = 48{,}190.75$$

$$C_o = 8 \times 48{,}190.75 = 385{,}526.04$$

Cat.	Option	Monthly cost (R$)	Months	Cost (R$)
1.1	AI safety and risk classification	48,190.75	8	385,526.04
1.1	Algorithmic fairness, non-discrimination, transparency, explainability and accountability	56,467.23	9	508,205.11
1.2	Controlled testing environments	51,102.14	12	613,225.68
1.2	Regulatory supervision	59,127.39	12 (ongoing)	709,528.68
1.3	AI procurement and vendor evaluation guidelines	29,144.03	5	145,720.14
1.3	Teacher and student AI-use policies	20,619.25	4	82,476.99
2.1	Data governance, protection and digital public infrastructure	57,642.83	12 (ongoing)	691,713.96
2.2	Secure hosting			missing amount

Cat.	Option	Monthly cost (R$)	Months	Cost (R$)
2.3	Open-source adoption and maintenance	33,128.49	12 (ongoing)	397,541.82
3.2	Training and professional development			missing amount
3.3	Governance committees	5,883.30	12 (ongoing)	70,599.60
3.3	Interministerial coordination	20,128.21	12 (ongoing)	241,538.52
	Direct cost, D			3,846,076.54
	Travel, 25% × D			961,519.14
	Overhead, 15% × (D + V)			721,139.35
	Administrative fee, 7% × (D + V)			336,531.70
	Known cost, C			5,865,266.73

With the three defaults, the additional costs add 52.5% to the direct cost. If the plan has no in-person activities, the institution already covers indirect costs and the funds are executed by a foundation charging 7%, the additional costs fall to R$269,225.36 (7% × D) and the known cost to R$4,115,301.90.

References

- ILO, ILOSTAT. Average monthly earnings of employees by sex, occupation and currency (DF_EAR_EMTA_SEX_OCU_CUR_NB). https://ilostat.ilo.org/data/
- IMF. World Economic Outlook: annual average consumer price inflation (PCPIPCH). Retrieved 9 November 2026. https://www.imf.org/external/datamapper/PCPIPCH@WEO
- SSA and ISSA. Social Security Programs Throughout the World, 2018–2019, table 4. https://www.ssa.gov/policy/docs/progdesc/ssptw/
- Lightcast. Beyond the Buzz: Developing the AI Skills Employers Actually Need, 2025. https://lightcast.io/resources/research/beyond-the-buzz-developing-the-ai-skills-employers-actually-need
- PwC. The Fearless Future: 2025 Global AI Jobs Barometer, 2025. https://www.pwc.com/gx/en/news-room/press-releases/2025/ai-linked-to-a-fourfold-increase-in-productivity-growth.html
- World Bank. Country classification by region and income group. https://api.worldbank.org/v2/country?format=json&per_page=400
- ExchangeRate-API. Reference rates with USD base, 3 October 2026. https://open.er-api.com/v6/latest/USD
- Global Fund. Data Explorer: budgets by cost category, cycles 5 to 7. https://data.theglobalfund.org/financial-insights
- United States. 2 CFR 200.414, indirect costs, 2024 revision. https://www.ecfr.gov/current/title-2/subtitle-A/chapter-II/part-200/subpart-E/subject-group-ECFRd93f2a98b1f6455/section-200.414
- UN Energy MPTF. 2025 annual financial report (7% indirect costs, UNSDG policy). https://mptf.undp.org/sites/default/files/documents/2026-05/2025_annual_financial_report-un_energy_mptf.pdf
- European Commission. AI Act proposal, COM(2021) 206, section 4. https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:52021PC0206
- Cambridge Forum on AI: Law and Governance. From watchdogs to partners in tech innovation: Norway's AI sandbox. https://www.cambridge.org/core/journals/cambridge-forum-on-ai-law-and-governance/article/from-watchdogs-to-partners-in-tech-innovation-navigating-regulatory-roles-in-norways-ai-sandbox/7E2FA0B1F75E6803B2DE4124EB0FC6CB
- Open Budget Kenya. Office of the Data Protection Commissioner, 2021–2025. https://openbudget.or.ke/project/1122002100_The_Office_of_the_Data_Protection_Commi/2024